

FINN 2041 CORPORATE FINANCE

2025/2026

Mergers & Acquisitions



Learning objectives

- ❑ By the end of the session, you should be able to:
 - Identify and discuss the motives for M&A that enhance shareholder value and those that do not
 - Outline the process of undertaking a M&A transaction
 - Evaluate the gains and costs that arise from a merger
 - Assess evidence on whether mergers and acquisitions create value and which stakeholders benefit
 - Discuss a variety of takeover defences

Required Readings

- Brealey, R., Myers, S., Allen, F. and Edmans, A. (2022) *Principles of Corporate Finance*, 14th ed., McGraw Hill: New York., Chapters 32

Introduction

☐ Merger: A transaction where two firms agree to integrate or combine their operations.

☐ Acquisition: A deal where a firm buys a controlling or 100% interest in another firm with the intent of making the acquired firm a subsidiary within its portfolio.

Introduction

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❑ Acquirer (or bidder) – the buyer of the firm

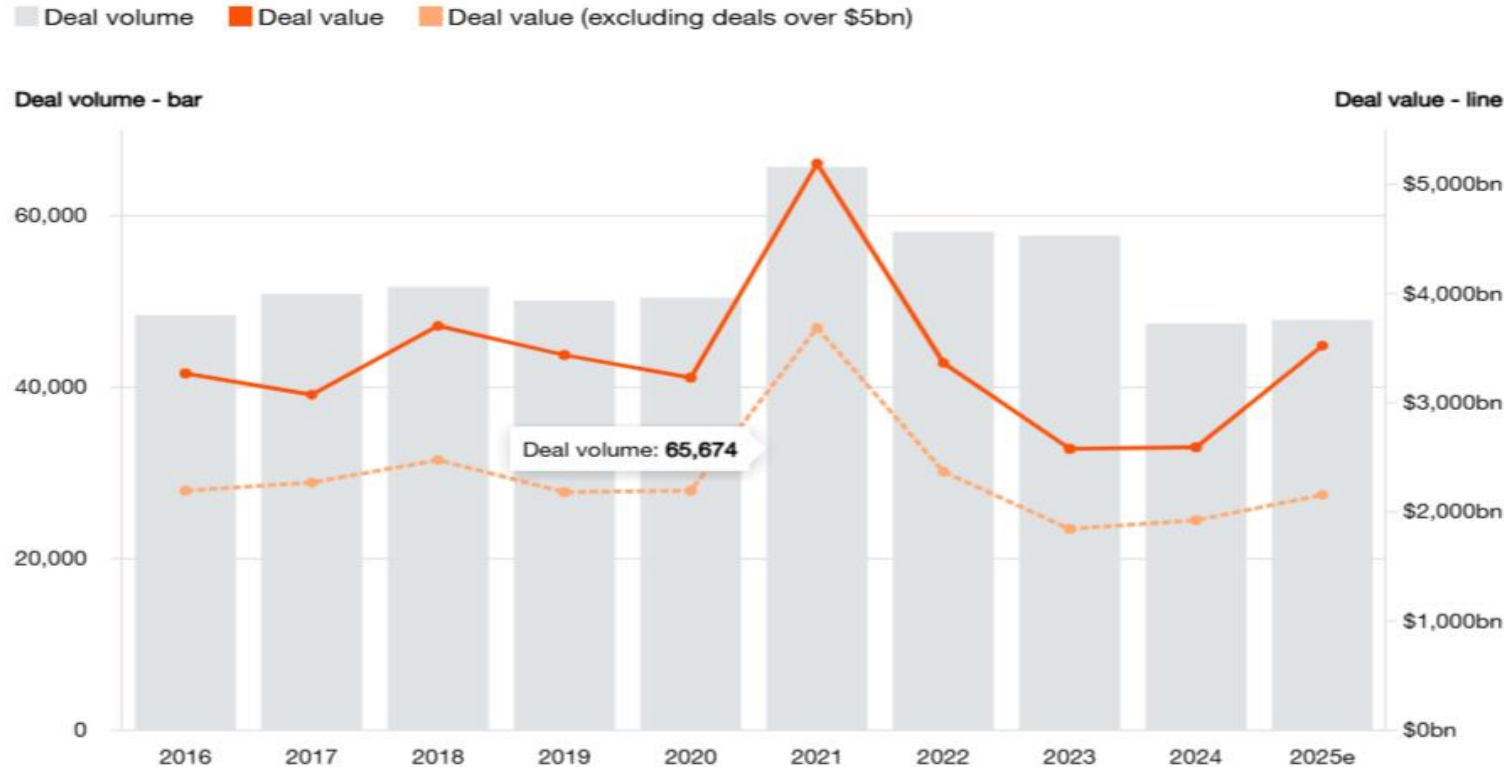
目标公司 被收购

❑ Target – Firm being acquired

❑ The global takeover market is highly active, averaging more than \$1 trillion per year in transaction value.

Introduction

Global M&A deal volumes and value (2016-2024)



Source: PWC

Introduction

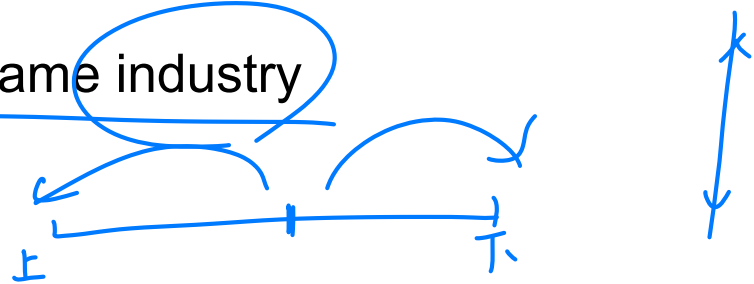
□ Types of Mergers

- Horizontal merger

- Merging companies are in the same industry
- E.g Exxon and Mobil in 1999

- Vertical merger

- Acquirer buys another company in the same production chain, i.e a supplier or distributor. Thus there is either a backward or forward integration
- Backward integration- Buying a supplier. E,g steel manufacturer buying an iron ore mining company
- Forward integration- Acquiring a distributor within the value chain- Iron ore mining company buying a steel producing company



Introduction

□ Types of Mergers

- Conglomerate merger
 - Acquirer purchases another company unrelated to its core business.
 - E.g General Electric acquisitions of many companies in different industries including media, finance and medical equipment.

Motives for Mergers and Acquisitions

□ Economies of Scale ←

- Cost savings enjoyed from producing goods in high volume that are not available to a small company

□ Economies of Scope ←

- Savings realized from combining the marketing and distribution of different types of related products

Motives for Mergers and Acquisitions

□ Expertise

- Particularly with new technologies, hiring experienced workers directly may be difficult.
- It may be more efficient to purchase the talent as an already functioning unit by acquiring an existing firm.

Motives for Mergers and Acquisitions

□ Monopoly gains

- Merging with or acquiring a major rival enables a firm to substantially reduce competition within the industry and thereby increase profits.
- However, most countries have antitrust laws that limit such activity.

Motives for Mergers and Acquisitions

□ Efficiency Gains

- Elimination of duplication
- Acquirers often argue that they can run the target organization more efficiently than existing management could.
- Although identifying poorly performing corporations is relatively easy, fixing them is another matter entirely

Motives for Mergers and Acquisitions

☐ Diversification

- Risk Reduction

- Like a large portfolio, large firms bear less unsystematic risk, so often mergers are justified on the basis that the combined firm is less risky.

- However, investors can achieve the benefits of diversification themselves by purchasing shares in the two separate firms

Motives for Mergers and Acquisitions

□ Diversification

- Debt Capacity and Borrowing Costs

- All else being equal, larger firms are more diversified and, therefore, have a lower probability of bankruptcy.
- Due to market imperfections like bankruptcy, a firm may be able to increase its debt and enjoy greater tax savings without incurring significant costs of financial distress by merging and diversifying.

Motives for Mergers and Acquisitions

□ Diversification

- Liquidity

- Shareholders of private companies often have a disproportionate share of their wealth invested in the private company.
- The liquidity that the bidder provides to the owners of a private firm can be valuable and often is an important incentive for the target shareholders to agree to the takeover.

Motives for Mergers and Acquisitions

□ Earnings growth

- It is possible for the earnings per share of the merged company to exceed the pre-merger earnings per share of either company, **even when the merger itself creates no economic value.**
- Merging a company with little growth potential and a company with high growth potential can raise earnings per share.
- However, the merger may add no economic value.

Managerial motives to merge

❑ Conflicts of Interest

- Managers may prefer to run a larger company due to additional pay and prestige.

❑ Overconfidence →

- Roll's "hubris hypothesis" maintains that overconfident CEOs pursue mergers that have low chance of creating value because they believe their ability to manage is great enough to succeed.

❑ Main difference

- Under the conflict of interest explanation, managers know they are destroying shareholder value but personally gain from doing so.
- Under the hubris hypothesis, managers believe they are doing the right thing for shareholders.

Cross-border motivations for M&A

❑ Exploiting market imperfections

eg. Taking advantage of differences in the relative cost of labour

❑ Overcoming adverse government policy

e.g to overcome protective tariffs, quotas or other barriers to free trade

❑ Technology transfer

acquiring a foreign company with superior technology in order to enhance acquirer's competitive position both home and abroad

❑ Following clients

To follow and support domestic clients more effectively

Motives for M&As – a warning

- ❑ Some motives/ sources of gains are quite valid such as operating synergies, eliminating inefficiencies, economies of scale and complementary resources.
- ❑ Other motives are questionable such as diversification and enhancing EPS.
- ❑ Investors and analysts have become more sceptical about potential gains from M&A and more aware of the potential downside of combining two firms.

The bidding process

□ Valuation

- The first step and the very key issue in a merger deal is to find out how much the target worth.
- Valuation is done mostly using:
 - Discounted cash flow (DCF) analysis
 - Comparable company analysis (price multiples) ←
 - Comparable transaction analysis- using recent takeover transactions for comparable companies to make direct estimates of the target's takeover value

The bidding process

❑ The Offer

- Once the acquirer has completed the valuation process, it is in the position to make an offer to purchase the company
- Not all offers are successful.
- Often, acquirers have to raise the price to consummate the deal (target's pre-bid market price **plus** the takeover premium)

The bidding process

❑ The Offer – Friendly vs Hostile

- Friendly- The board of the target company agrees the terms with the bidder, and then recommends it to the shareholders.
- Hostile- The target's managers may resist the merger offer (price too low or simply want to keep their jobs).
 - The acquirer may therefore approach the target's shareholders directly to make an offer to purchase firm at a specific price

OR

- Proxy fight- seeking to replace board via shareholder vote

The bidding process

❑ The Offer - Method of payment

1. Cash - the bidder simply pays for the target, including any premium, in cash.

- Attractive to target shareholders as compensation they receive for their shares is certain in value.
- Target shareholders may, however, be liable to pay capital gains tax on shares sold to bidder.
- For bidder, can see exactly how much is being offered.
- No effect on number of issued shares in bidder firm.

The bidding process

❑ The Offer - Method of payment

acquirer → target

2. Stock-for-stock - the bidder pays for target by issuing new stock and giving it to the target shareholders.

• Exchange Ratio

The number of bidder shares received in exchange for each target share

- A stock-swap merger is a positive-NPV investment for the acquiring shareholders if the share price of the merged firm exceeds the pre-merger price of the acquiring firm.

The bidding process

□ The Offer - Method of payment

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2. Stock-for-stock

Let A be the pre-merger value of the acquirer, T be the pre-merger value of the target, and S be the value of the synergies created by the merger. If the acquirer has N_A shares outstanding before the merger and issues x new shares to pay for the target, then the acquirer's share price should increase post-acquisition if

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每股 = $V_{\text{合}} \leftarrow \frac{A + T + S}{N_A + x}$

合并后的每股价格.

收购前

acquire 公司 收购前

$\frac{A}{N_A} = P_A$

1000

The bidding process

❑ The Offer - Method of payment

2. Stock-for-stock

Advantages for target company shareholders:

- Retain equity interest in their company.
- No brokerage costs from reinvesting cash and no capital gains tax liability.

Disadvantages to acquiring company and its shareholders:

- Increases number of shares in circulation.
- Share issue may move bidder away from its optimal capital structure.

debt-to-equity ↑ ratio

The bidding process

□ The Offer - Method of payment

3. Mixed bid

- A stock-for-stock offer is supported by a cash alternative.
- Increasingly more popular financing choice because:
 - they are seen as being more acceptable to the target company's shareholders

The bidding process

❑ Approval: Shareholders

- For a merger deal to proceed, both the target and acquiring board of directors must approve the deal and put the question to a vote of the shareholders of the target (and, in some cases, the shareholder of the acquiring firm as well).

❑ Approval: Market Regulators

- All mergers must be approved by regulators.
- In most countries all M & A transaction above a certain size must be approved by the government before the proposed takeovers occur.

Estimating merger gains and costs

□ NOTATION

Let:

V_A = Pre-merger value of acquirer

V_T = Pre merger value of target

V_{AT} = Post merger value of combined firm

C = Cash paid for the target

Estimating merger gains and costs

- ❑ Synergies (Incremental value)- Value of combined firm greater than the sum of separate values.

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$$\text{Synergies} = V_{AT} - (V_A + V_T - C)$$

→ Thus, $V_{AT} = V_A + V_T + \text{Synergies} - C$

- ❑ NPV of merger (Gain to acquirer)

$$\text{NPV} = \text{Synergies} - (C - V_T)$$

$$C - V_T = \text{Premium}$$

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$C - V_T$ is also the gain to target

Estimating merger gains and costs

□ Example

Gobi Desserts is bidding to take over Universal Puddings. Gobi has 3,000 shares outstanding, selling at £50 per share. Universal has 2,000 shares outstanding, selling at £17.50 a share. Gobi estimates the economic gain from the merger to be £15,000.

syn

Estimating merger gains and costs

- a. If Universal can be acquired for £20 a share, what is the NPV of the merger to Gobi?

↑ C 10% →

$$\text{NPV} = \text{Synergies(Incremental value)} - \text{Premium paid}$$

$$\text{Synergies(Incremental value)} = 15,000$$

$$\text{Premium} = (C - V_T) = (£20 \times 2,000) - (£17.50 \times 2,000) = £5,000$$

$$\text{NPV} = 15,000 - 5,000 = £10,000$$

Estimating merger gains and costs

b. Suppose that the merger takes place through an exchange of stock. On the basis of the pre-merger prices of the firms, Gobi sells for £50, so instead of paying £20 cash, Gobi issues 0.40 of its shares for every Universal share acquired. What will be the stock price of the merged firm?

Value of firm =

N of stocks = 原 + 新发行的

$$= \frac{10 \times 3K + 17.5 \times 2K + 15000 - 2K \times 0.4 \times 50}{3K + 800} = \frac{V_A + V_T + S_{yn} - C}{V_A + V_T + S_{yn} - C}$$

Estimating merger gains and costs

$$V_{\text{Combined firm}} = V_{\text{Gobi}} + V_{\text{Universal}} + \text{Synergies(Incremental value)}$$

$$= (3,000 \times 50) + (2,000 \times 17.5) + 15,000 = \text{£}200,000$$

Shares issued to B shareholders is $2,000 \times 0.4$ or 800 shares.

$$\text{Share Price}_{\text{Combined Firm}} = \text{£}200,000 / (3,000 + 800) = \text{£}52.63$$

Handwritten notes: "50" above the 3,000, and blue circles around "800" and "£52.63" with an arrow pointing to the result.

Estimating merger gains and costs

c. What is the NPV of the merger to Gobi when it uses an exchange of stock?

To find the NPV in a stock offer, we compare how what the shareholders of the acquiring firm are giving up to acquire their rights in the new company.

$$NPV = -\text{成本} + \text{获得}$$

$$-150000 \quad 157800$$

50\$ Shareholders of Gobi own 79% of the combined firm (3000 shares / 3800 shares)

800 → target firm

VAT.

This translates to an equity value of £157,800 = 0.79 x £200,000

Thus, Gobi shareholders are giving their ownership in the old firm (£150,000) to gain £157,800 in the new firm.

The NPV is therefore = £157,800 - £150,000 = £7,800

Shareholder wealth effects of M&A

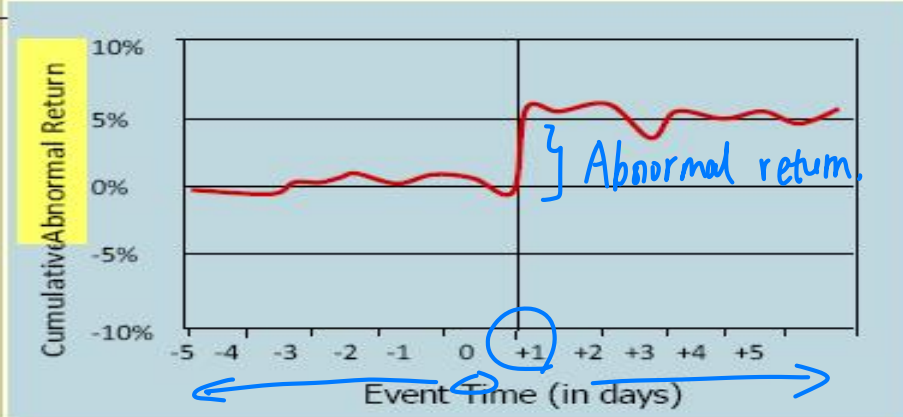
- ❑ How do M&As create wealth for shareholders in terms of increases in share prices around such activities?
- ❑ **Event studies** analysing short-term shareholder wealth effects constitute the dominant approach for measuring wealth effects since the 1970s.
- ❑ The event study methodology assumes that an M&A announcement brings new information to the market.
 - Hence, investors' expectations about the firm's prospects are updated and reflected in the share prices.

Shareholder wealth effects of M&A

- ❑ An **abnormal return** equals the difference between the realised returns and an expected (benchmark) return,
 - Common benchmarks are estimated using asset pricing models such as the market model, CAPM, or the Fama and French 3-factor model.
- ❑ This methodology can also be applied to assess the long-term shareholder wealth effects.

Shareholder wealth effects of M&A

Event Studies



Abnormal return

The actual return minus the
expected return

Could just be the market index
return for the day, or the
market index return times the
beta of the firm reporting the
earnings announcement

The positive bump on day 0 implies that the earnings news was, on average for these firms, better than expected!

Because the line is flat after day 0, this means that the market fully incorporated the earnings news on the event day...no additional upward or downward price trend is seen.

Shareholder wealth effects of M&A

□ Short-run evidence

- Overall, the shareholders of target firms accumulate *significant* positive returns in the period around the bid announcement.
- Overall, shareholders of the bidding firm earn *insignificant* abnormal returns prior to and at the announcement of a takeover.
 - E.g. Studies of late 1970s and 1980s: Morck et al. (1990); Byrd and Hickman (1992), and Chang (1998): *negative* abnormal returns between -1.2% to -0.7%.

Shareholder wealth effects of M&A

□ Short-run evidence

Dodd (1980), "Merger proposals, management discretion and stockholder wealth,"
Journal of Financial Economics, Volume 8, Issue 2, June 1980, Pages 105-137

- 151 targets and 126 bidders over 1970-1977
- * AR is Abnormal Return = Actual – Expected. Reported AR is average of firm ARs.

		Bidders	Targets
Successful	2-day AR *	-1.09%	13.41%
	Sample Size	60	71
	T-statistics	-3.0	23.8
Unsuccessful	2-day AR	-1.24%	12.73%
	Sample Size	66	80
	T-statistics	-2.6	19.1

Shareholder wealth effects of M&A

- ❑ Why do sellers (targets) typically earn high(er) returns?
 - Evidence suggests that the premium the acquirer pays is approximately equal to the value it adds, which means the *target shareholders* ultimately capture the value added by the acquirer.
 - It is often the case that there is competition among bidders, which increases the gains for the target firm == >> directly results in large takeover premiums.

Shareholder wealth effects of M&A

□ Long-run effect

- This implies that we should assess the wealth effects from acquisitions using long-run windows.
- Windows usually used: 1,2,3,5 years after the merger.
- Some problems associated with long run windows:
 - Other events occurring within this window will impact on the valuation of the merger: Strategic, Policy changes, Other acquisitions...etc. reduced reliability of long term results

Shareholder wealth effects of M&A

□ Long-run effect

- The majority of long-run studies show that acquisitions are not value-creating.
- The evidence on long-term abnormal returns demonstrates that takeovers lead to a decline in share prices over several months/years subsequent to the transaction.
- However, the results may be sensitive to the method of estimation.

Summary of empirical research

❑ Shareholders:

- Accounting studies indicate that acquisitions are unprofitable to shareholders of acquirer.
- Gain of target company shareholders' is likely to be a result of bid premium.
- Acquiring company shareholders' lack of gain may be due to high anticipation of acquisition in developed market. May also be due to overpayment, managerial hubris.

Summary of empirical research

❑ Shareholders:

- But some more recent studies are beginning to document slight gains to acquiring companies (see Alexandrilidis et al (2017) and Tunyi (2021))
 - Alexandridis, G., Antypas, N., & Travlos, N. (2017). Value creation from M&As: New evidence. *Journal of Corporate Finance*, 45, 632-650.
 - Tunyi, A. A. (2021). Revisiting acquirer returns: Evidence from unanticipated deals. *Journal of Corporate Finance*, 66, 101789.

Summary of empirical research

❑ Managers and employees:

- Acquiring company managers benefit from:
 - increased power and status;
 - increased financial rewards;
 - increased job security.
- Managers of target companies tend to lose their jobs following an acquisition, while their employees also face uncertain futures.

Summary of empirical research

❑ The winners:

- Acquiring company managers, financial institutions and target company shareholders

❑ The losers:

- Acquiring company shareholders, target company managers and employees

Takeover defences

Mechanisms adopted by Target to prevent a takeover either before or after it is initiated

□ Before initiation (Deterrents)

1. Poison Pills

- It is a rights offering that gives the target shareholders the right to buy shares in either the target or an acquirer at a deeply discounted price.
- Because target shareholders can purchase shares at less than the market price, existing shareholders of the acquirer effectively subsidize their purchases, making the takeover unattractive

Takeover defences

2. Golden Parachute

- An extremely lucrative severance package that is guaranteed to a firm's senior management in the event that the firm is taken over and the managers are let go

Takeover defences

3. Staggered Board (Classified Board) 交错.

- In many public companies, a board of directors three-year terms are staggered so that only one-third of the directors are up for election each year.
- A bidder's candidate would have to win a proxy fight two years in a row before the bidder had a majority presence on the target board.

Takeover defences

4. Supermajority Voting Provisions 100%

- Changing charter and bylaws to provide for a higher percentage approval by shareholders for mergers than normally is required. A typical provision might require a vote of 80 percent of the outstanding shares of the target company (as opposed to a simple 51 percent majority).

5. Fair price amendments

- changes to the corporate charter and bylaws that disallow mergers for which the offer is below some threshold. E.g example, requiring an acquirer to pay at least as much as the highest stock price at which the target has traded in the public market over a specified period. This protects targets against temporary declines in their share prices by setting a floor value bid.

Takeover defences

☐ After initiation

1. “Just say no”

- Managers simply decline the offer. Acquirer may increase the bid or reveal its strategy to advance the takeover.

2. Litigation

- Filing a law suit alleging some violations of securities or antitrust laws. This approach mostly ends up as a delay tactic.

Takeover defences

3. White Knight ✓

- A target company's defense against a hostile takeover attempt in which it looks for another, friendlier company to acquire it

4. White Squire ✓

- A variant of the white knight defense, in which a large, passive investor or firm agrees to purchase a substantial block of shares in a target with special voting rights

Takeover defences

5. Greenmail ←

- An agreement allowing the target to repurchase its own shares back from the acquiring company, usually at a premium to the market price. Usually accompanied by an agreement that the acquirer will not pursue another hostile takeover attempt of the target for a set period.

6. Crown Jewel defence →

- A target may decide to sell off a subsidiary or asset to a third party. This may cause the acquirer to abandon the takeover if the acquisition of this asset was the acquirer's major motivation.

Takeover defences

7. Recapitalisation

- With recapitalization, a company changes its capital structure to make itself less attractive as a target.
- For example, companies might choose to issue debt and then use the proceeds to repurchase stock.

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